

Amulet Motion Controller

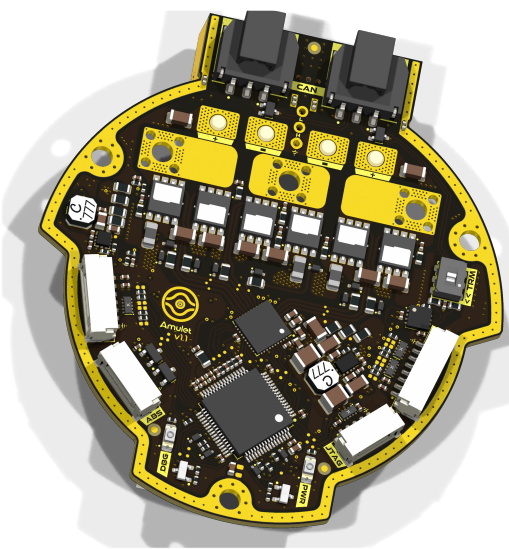
Variant:

2026-08-17

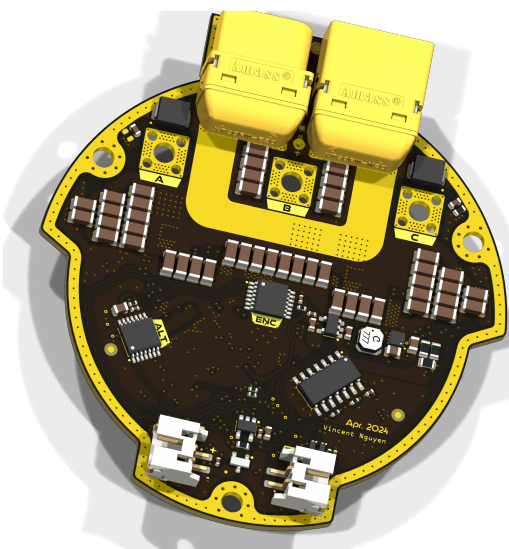
Rev 1.1.1+ (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 17-Aug-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

	Comments:	Company: EPFL Xplore Research	Variant:	Git Hash: 5b68472
	Sheet Title: Cover Page	Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Path: /	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13
			Revision: 1.1.1+ (Unreleased)	
			Size: A3	Sheet: 1 of 21

[2] Block Diagram

Page 4 (Power)
Page 5 (I/Os)

Page 6

+VBAT

Buck Max 600 mA

+12V

+5V

Buck Max 600 mA

~225 uF Bulk Capacitance

+5V

Buck Max 500 mA

+3V3

LDO Max 250 mA

+A3V3

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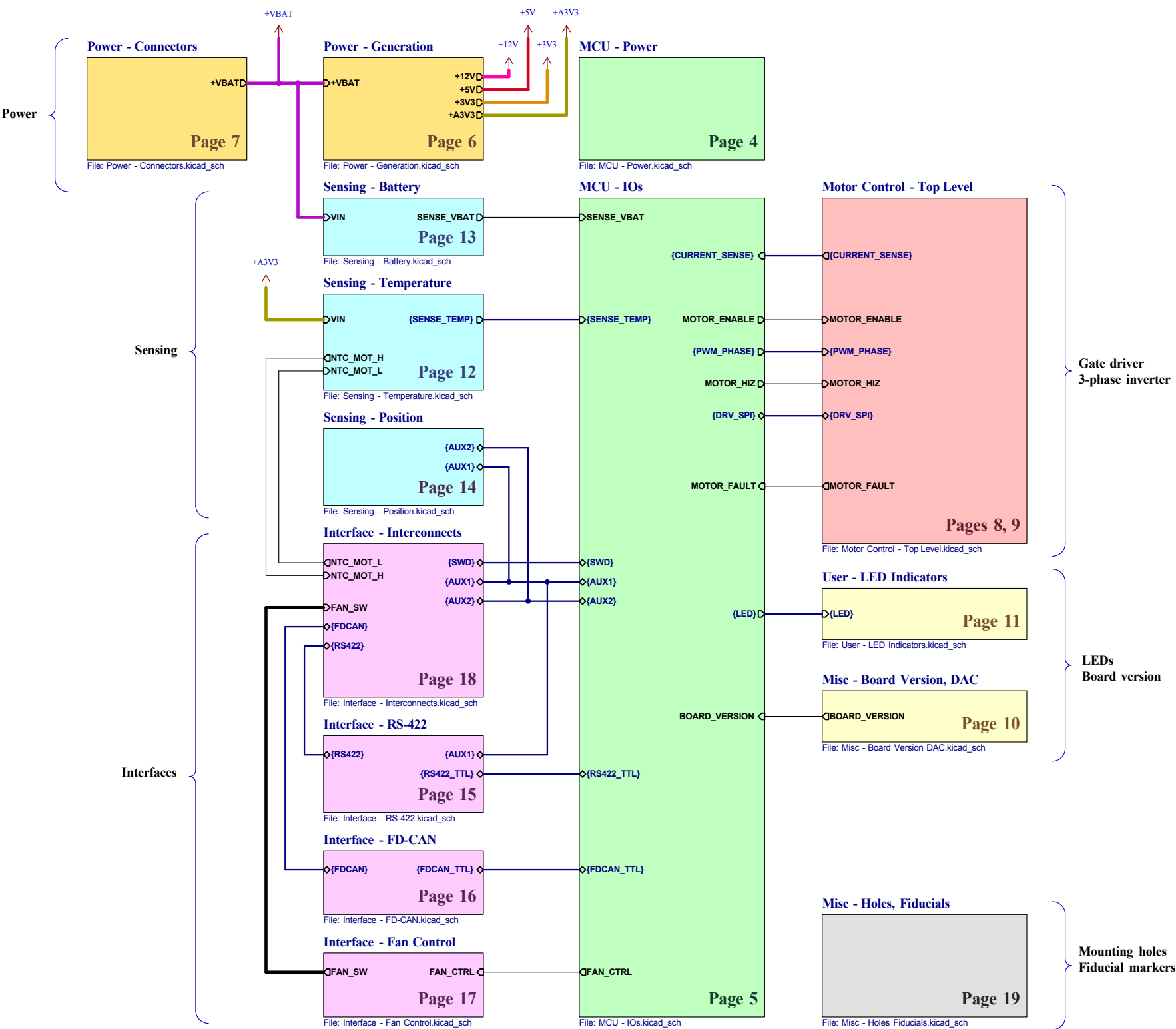
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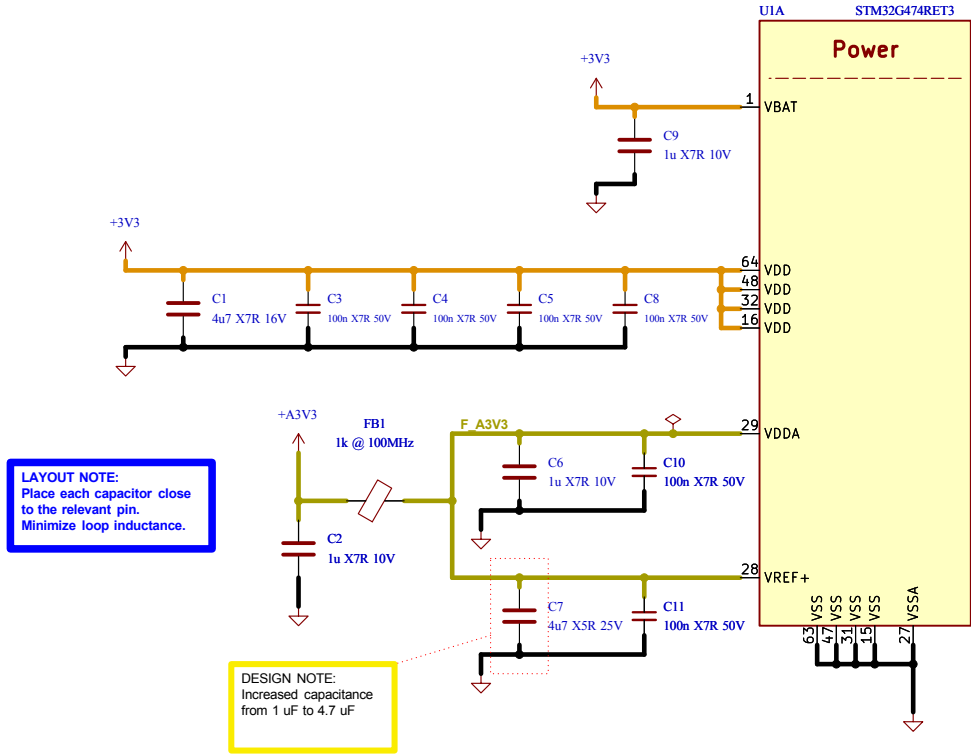
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	EPFL Xplore Research		5b68472	
	Board Name:	Project Name:		
	Amulet Motion Controller	Chienpanzé		
Sheet Title:	File Name:	Designer:	Date:	Revision:
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Sheet Path:	Reviewer:	Size:	Sheet:	
/Block Diagram/		A3	2 of 21	

[3] Project Architecture



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	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1.1+ (Unreleased)
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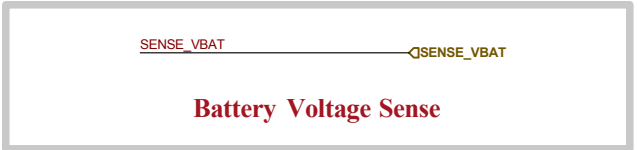
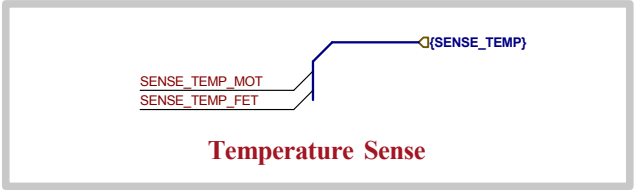
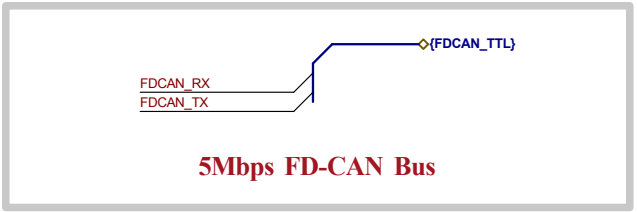
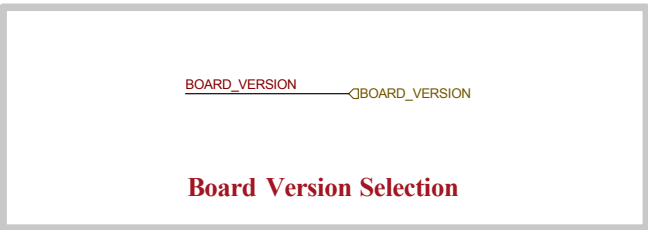
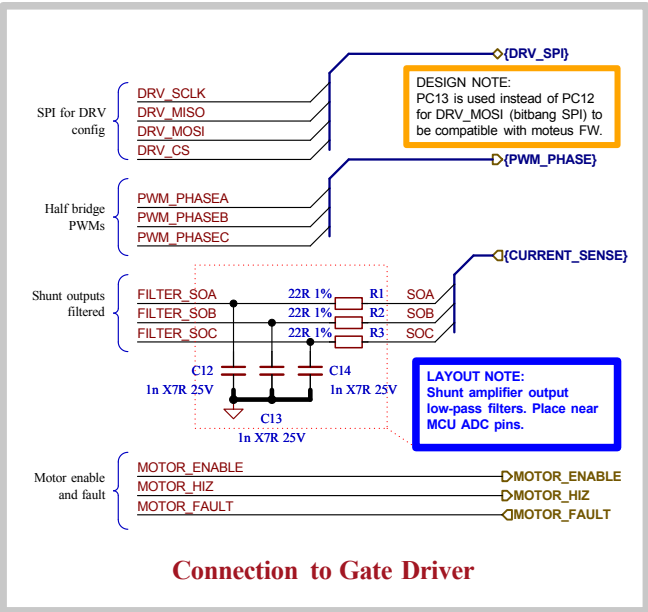
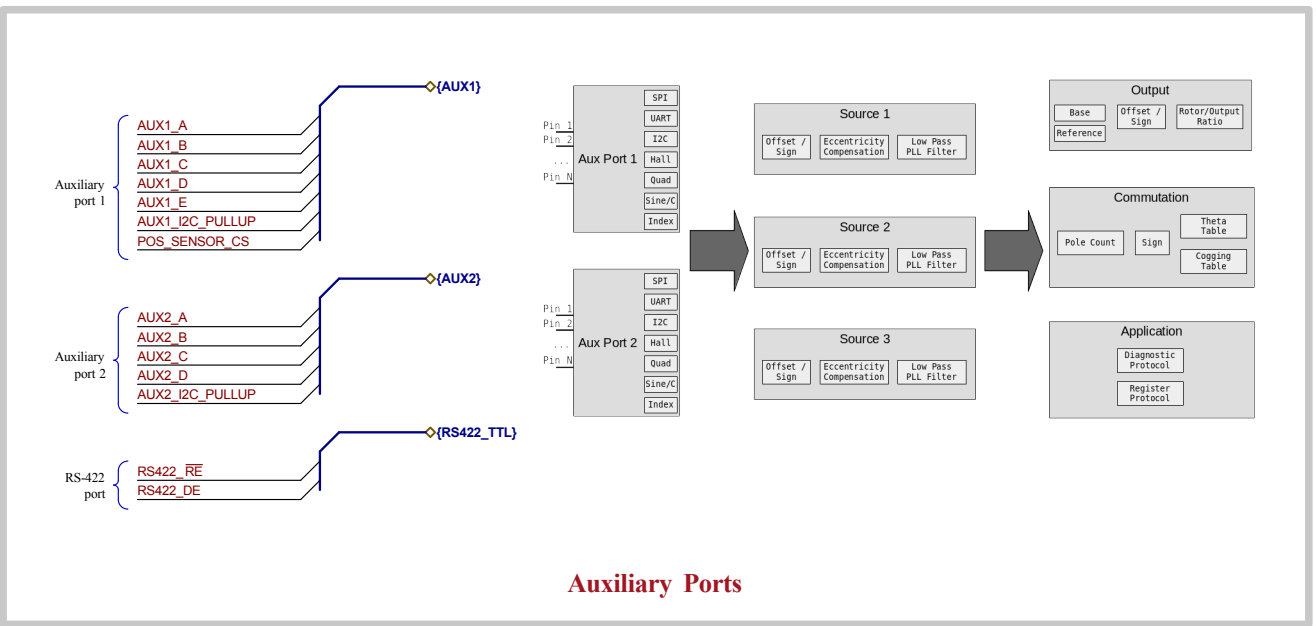
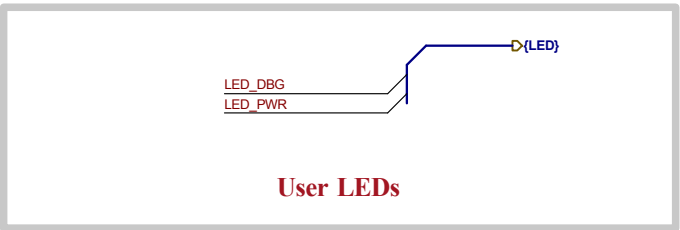
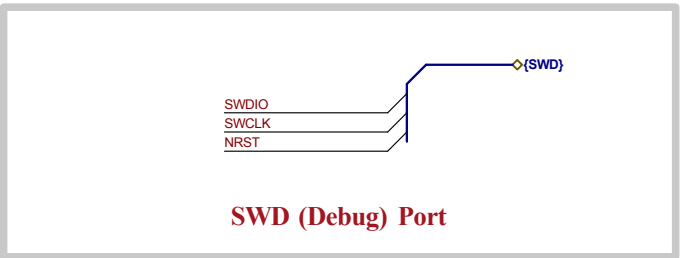
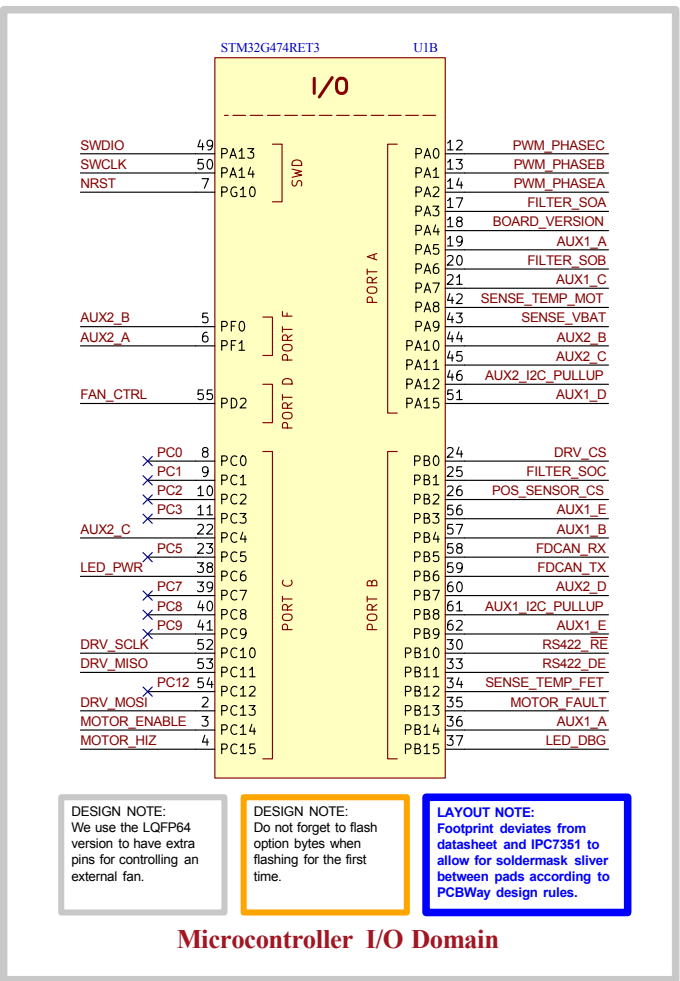
[4] MCU - Power



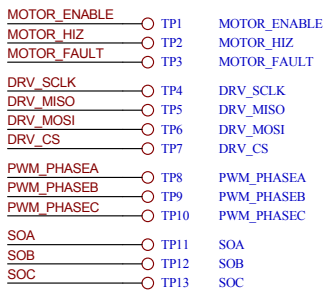
Microcontroller Power Supply Scheme

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	Sheet Title: MCU - Power	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Path: /Project Architecture/MCU - Power/	File Name: MCU - Power.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-18	Revision: 1.1.1+ (Unreleased)
			Reviewer:	Size: A4	Sheet: 4 of 21

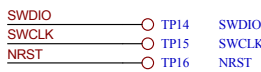
[5] MCU - I/Os



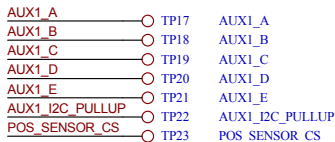
Gate Driver



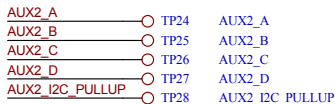
Debug



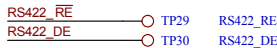
Auxiliary pins 1



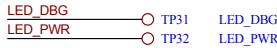
Auxiliary pins 2



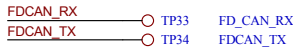
RS-422



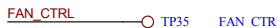
LEDs



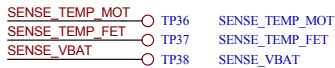
FD-CAN



Fan



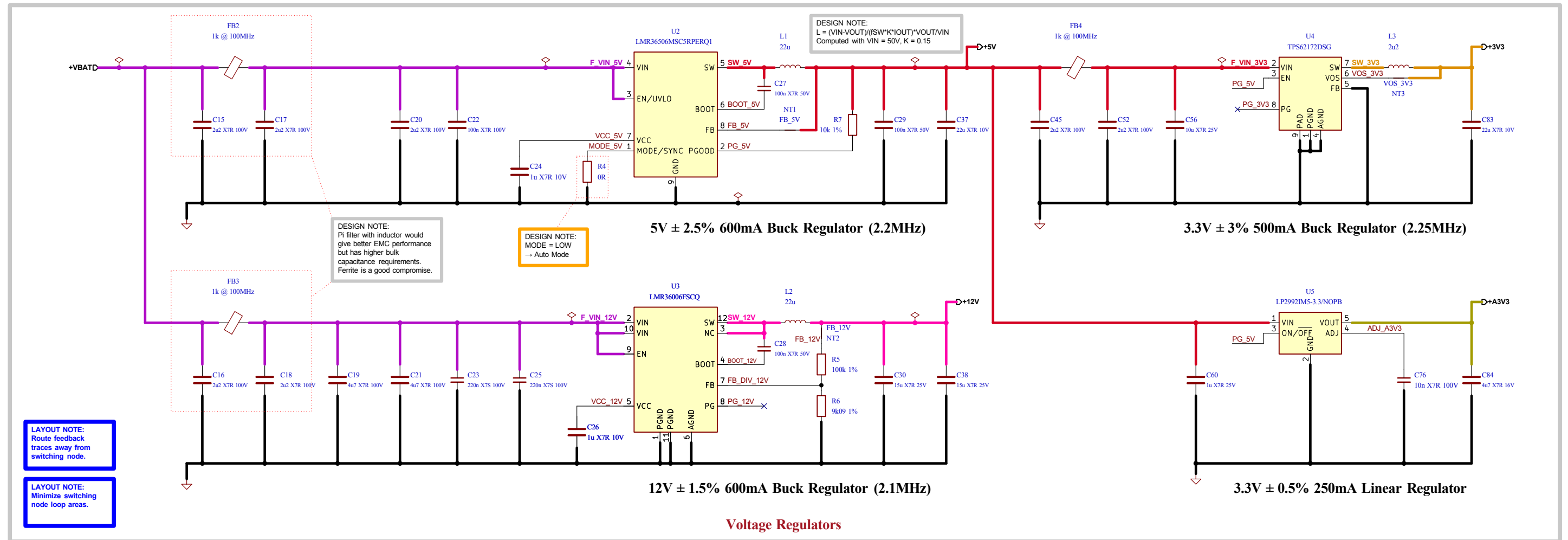
Sense



Test Points

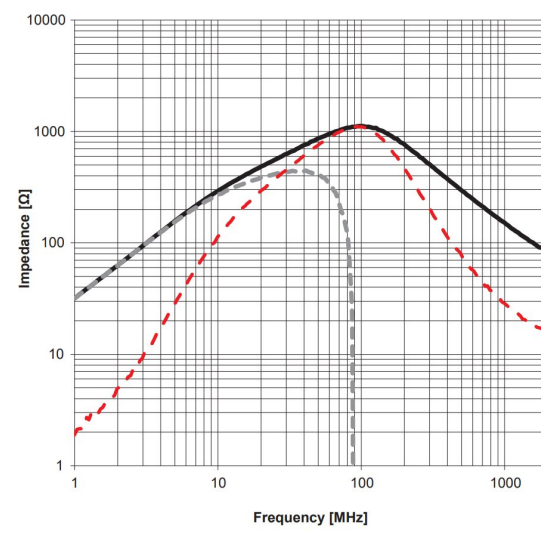
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	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: MCU - I/Os	File Name: MCU - IOs.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-20
Sheet Path: /Project Architecture/MCU - IOs/	Reviewer:	Size: A3	Revision: 1.1.1+ (Unreleased)	Sheet: 5 of 21

[6] Power - Generation



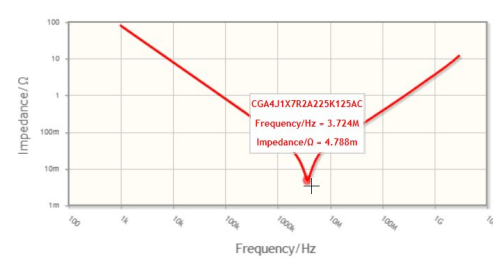
Voltage Regulators

Ferrite bead Impedance vs Frequency curve

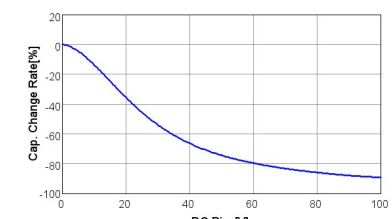


Capacitor and Ferrite Bead Parameters

Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



Battery

+VBAT TP39 +VBAT
+VBAT TP40 +VBAT
+VBAT TP41 +VBAT

Filtered Power Rails

F_VIN_12V TP42 F_VIN_12V
F_VIN_5V TP43 F_VIN_5V
F_VIN_3V3 TP44 F_VIN_3V3

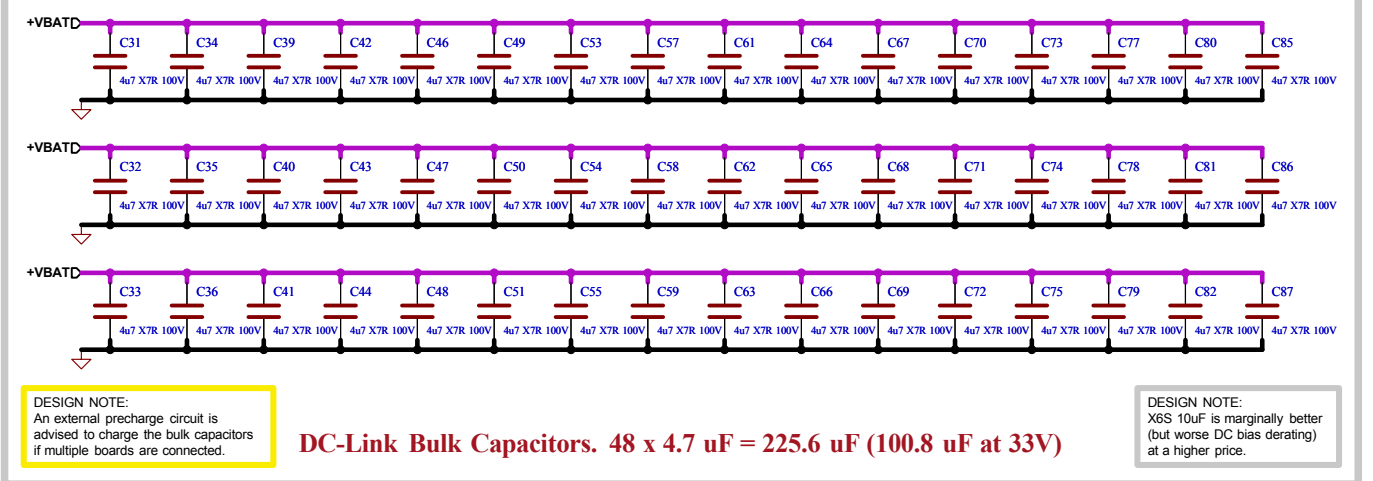
Power Rails

+12V TP45 +12V
+5V TP46 +5V
+5V TP47 +5V
+3V3 TP48 +3V3
+3V3 TP49 +3V3
+3V3 TP50 +3V3
+A3V3 TP51 +A3V3

Ground

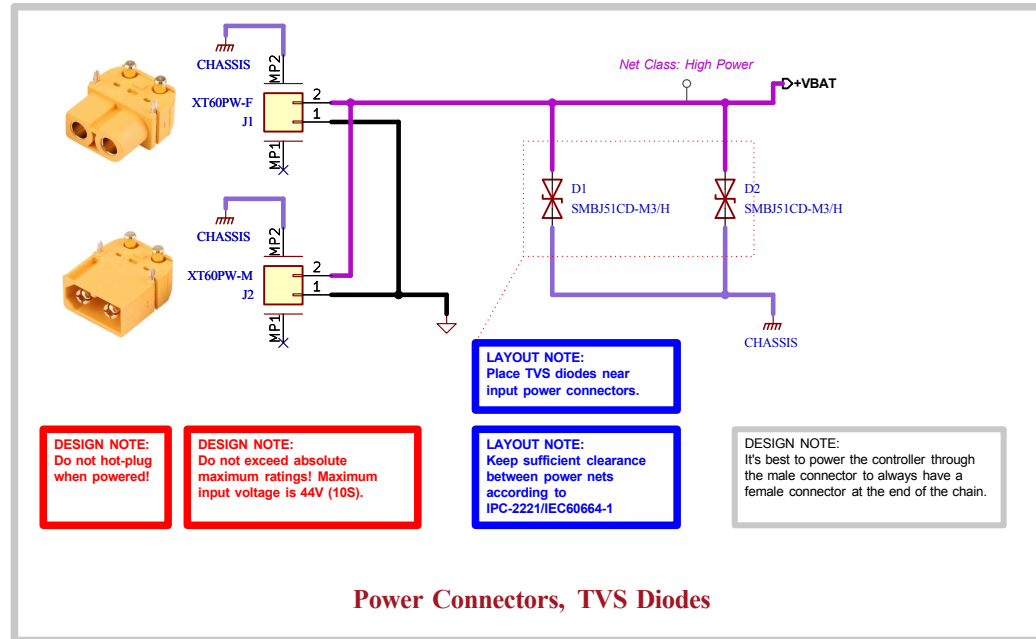
GND TP52 GND
GND TP53 GND
GND TP54 GND
GND TP55 GND
GND TP56 GND
GND TP57 GND
GND TP58 GND

Test Points



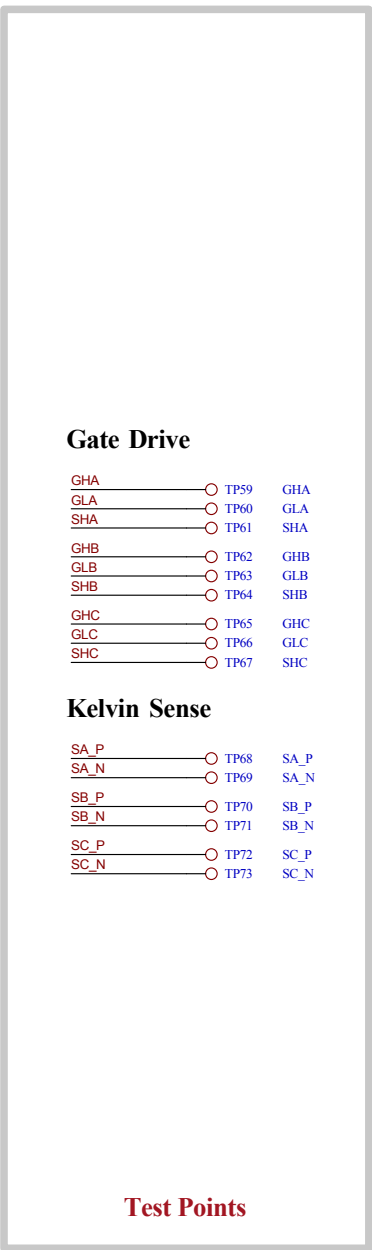
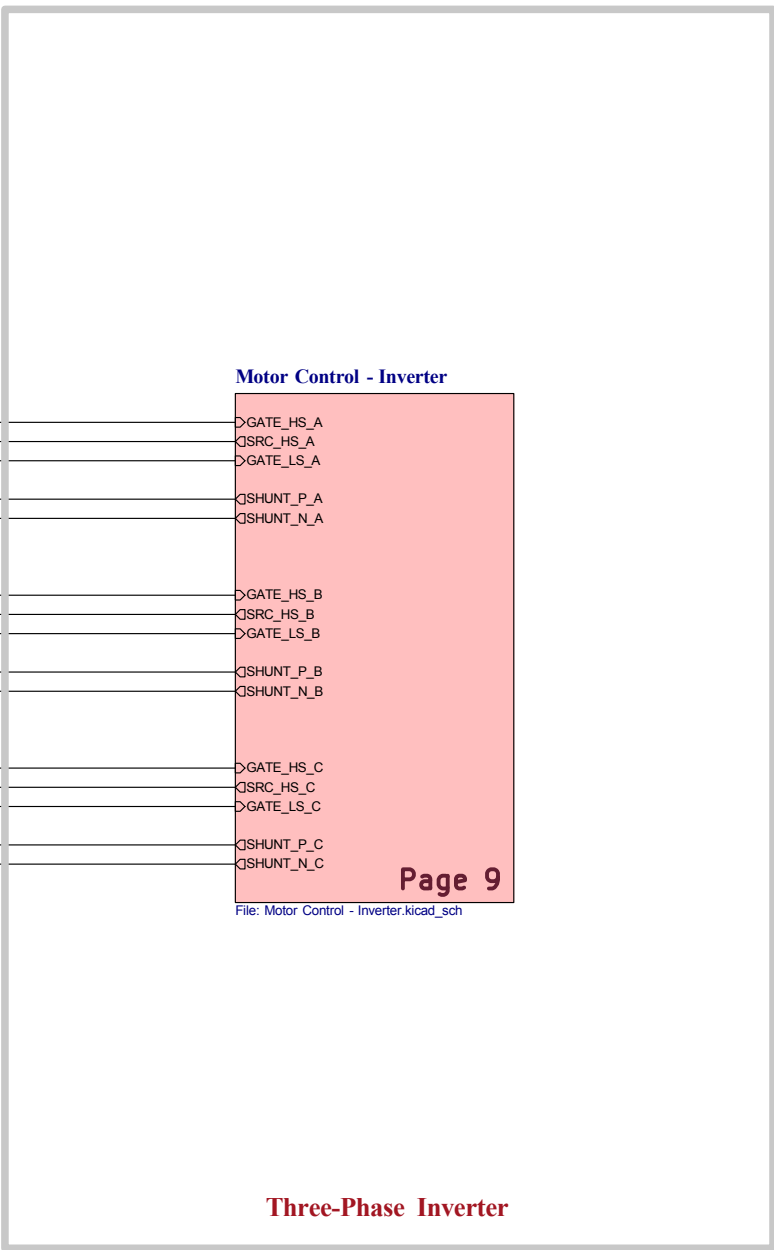
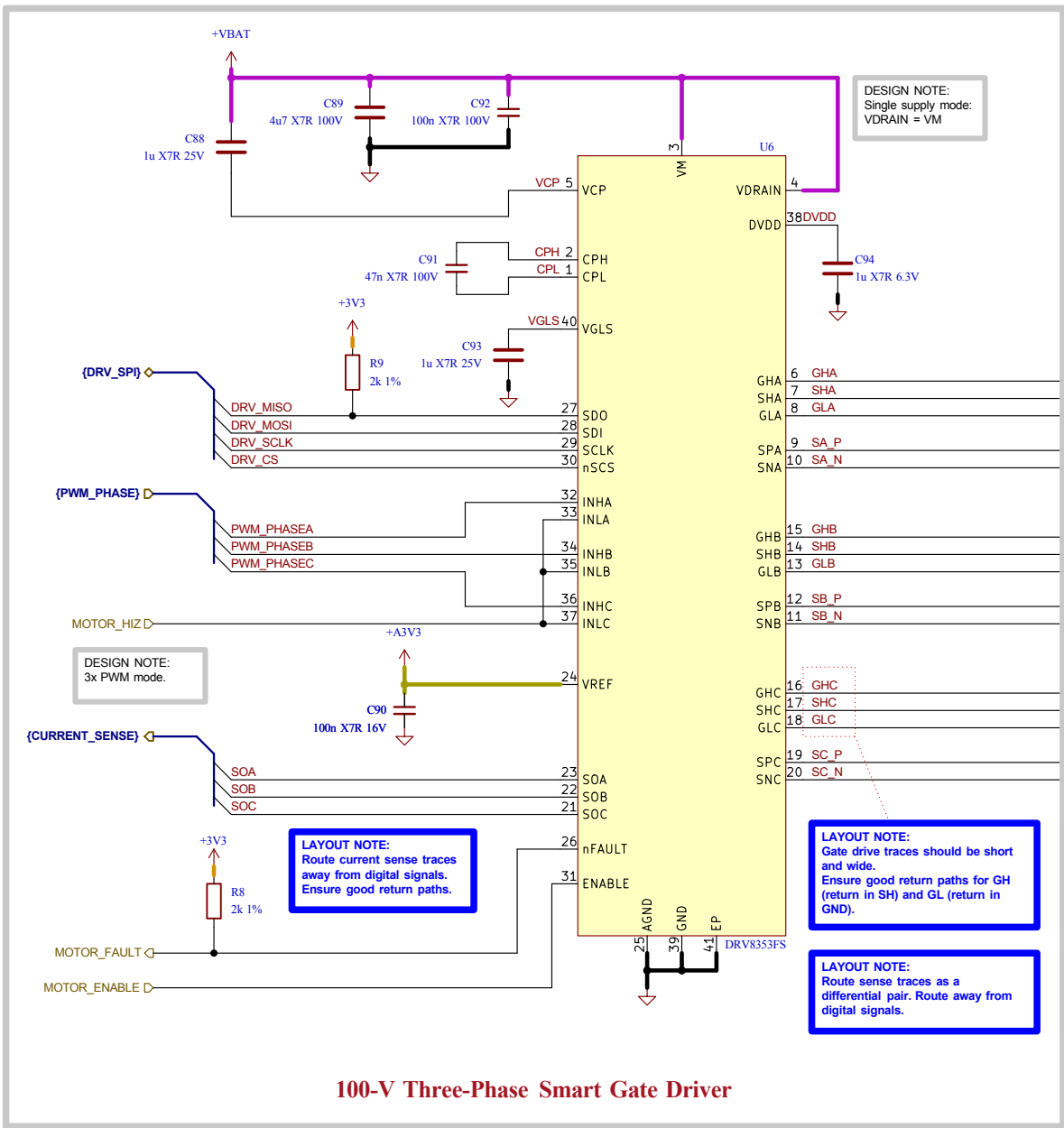
Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472	
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé		
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)	
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: A3	Sheet: 6 of 21	

[7] Power - Connectors



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 7 of 21

[8] Motor Control - Top Level



Motor Control - Inverter

Gate Drive

GHA	TP59	GHA
GLA	TP60	GLA
SHA	TP61	SHA
GHB	TP62	GHB
GLB	TP63	GLB
SHB	TP64	SHB
GHC	TP65	GHC
GLC	TP66	GLC
SHC	TP67	SHC

Kelvin Sense

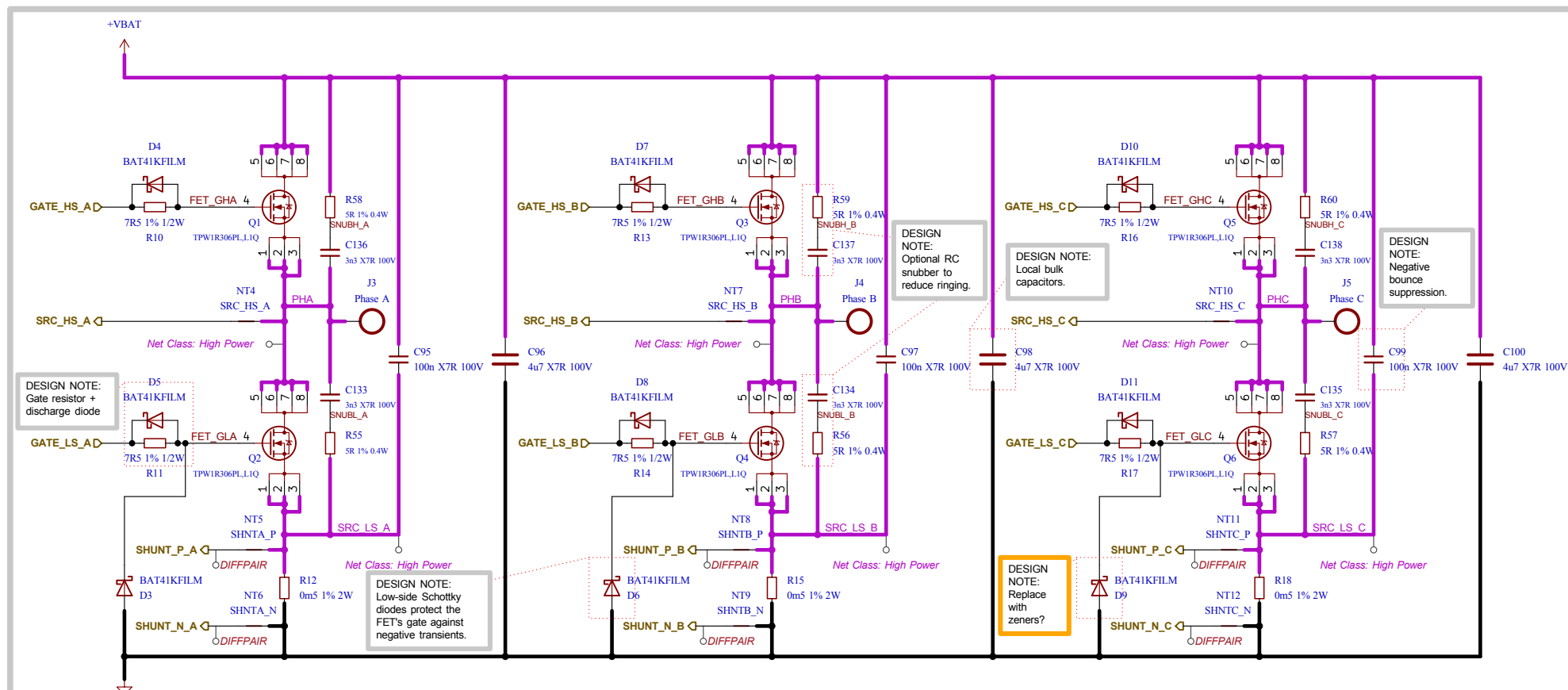
SA_P	TP68	SA_P
SA_N	TP69	SA_N
SB_P	TP70	SB_P
SB_N	TP71	SB_N
SC_P	TP72	SC_P
SC_N	TP73	SC_N

Page 9

File: Motor Control - Inverter.kicad_sch

	Comments:	Company: EPFL Xplore Research	Variant:	Git Hash: 5b68472
	Sheet Title: Motor Control - Top Level	Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Path: /Project Architecture/Motor Control - Top Level/	File Name: Motor Control - Top Level.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-20
			Revision: 1.1.1+ (Unreleased)	
			Size: A3	Sheet: 8 of 21

[9] Motor Control - Inverter



Three-Phase Inverter

LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance
between power nets according
to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:

System Design Considerations for
High-Power Motor Driver Applications

Best Practices for Board Layout
of Motor Drivers

Proper RC Snubber Design
for Motor Drivers

Sheet Title:

Motor Control - Inverter

Sheet Path:

/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:

EPFL Xplore Research

Board Name:

Amulet Motion Controller

File Name:

Motor Control - Inverter.kicad sch

Designer:

Vincent Nguyen

Reviewer:

100

Variant:

Project Name:	
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Chienpanzé

Date:

2024-01-25

	Git Hash:
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5b68472

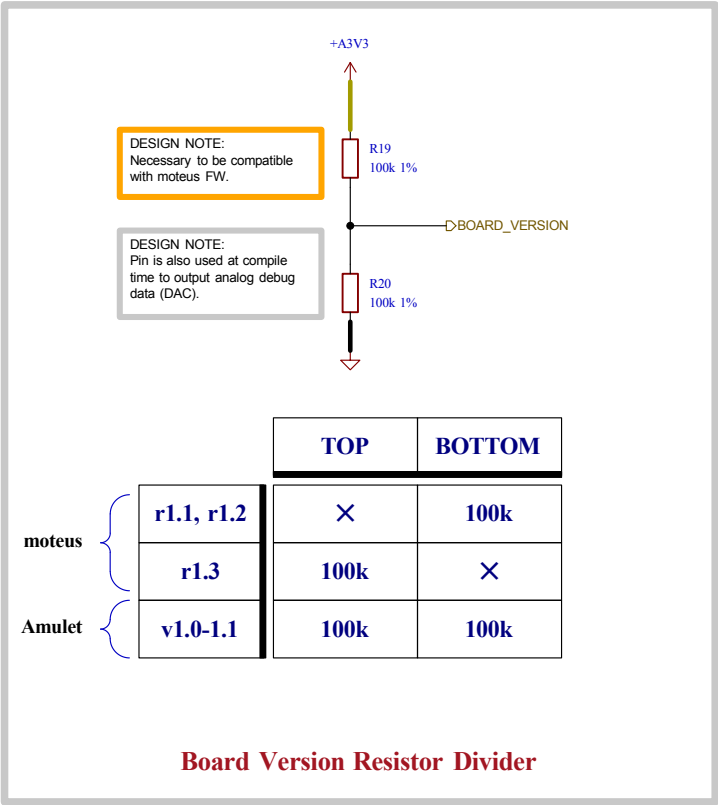
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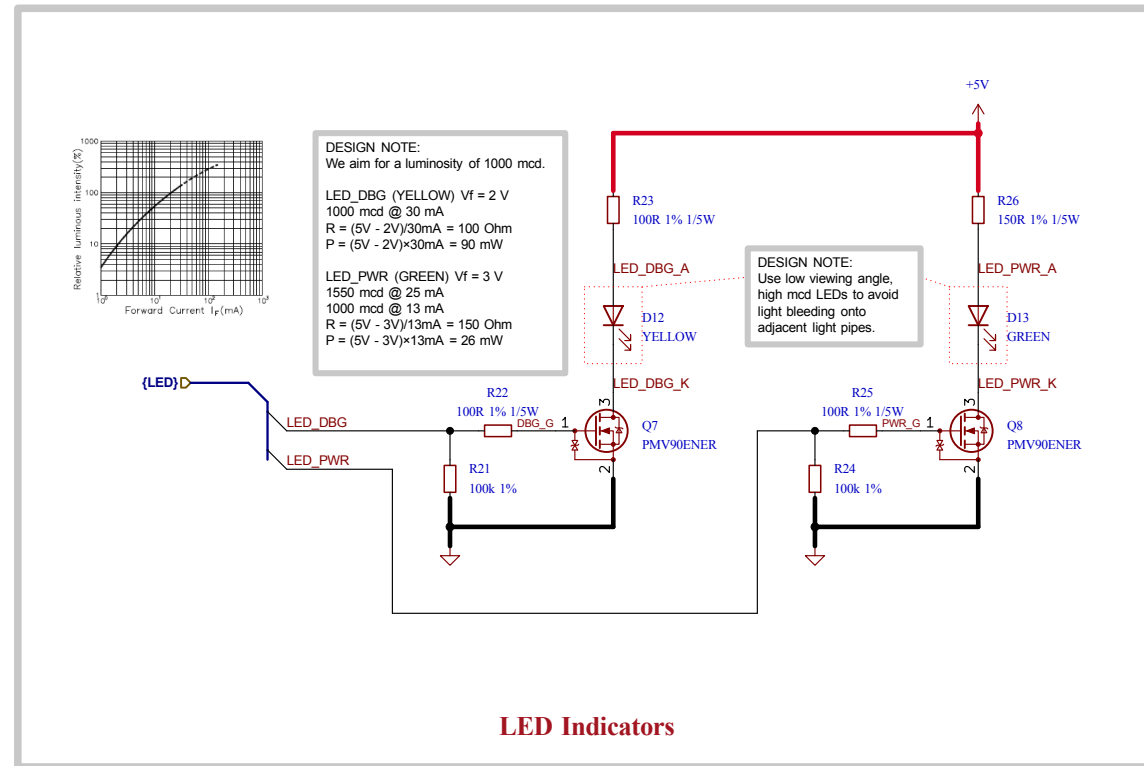
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[10] Misc - Board Version, DAC



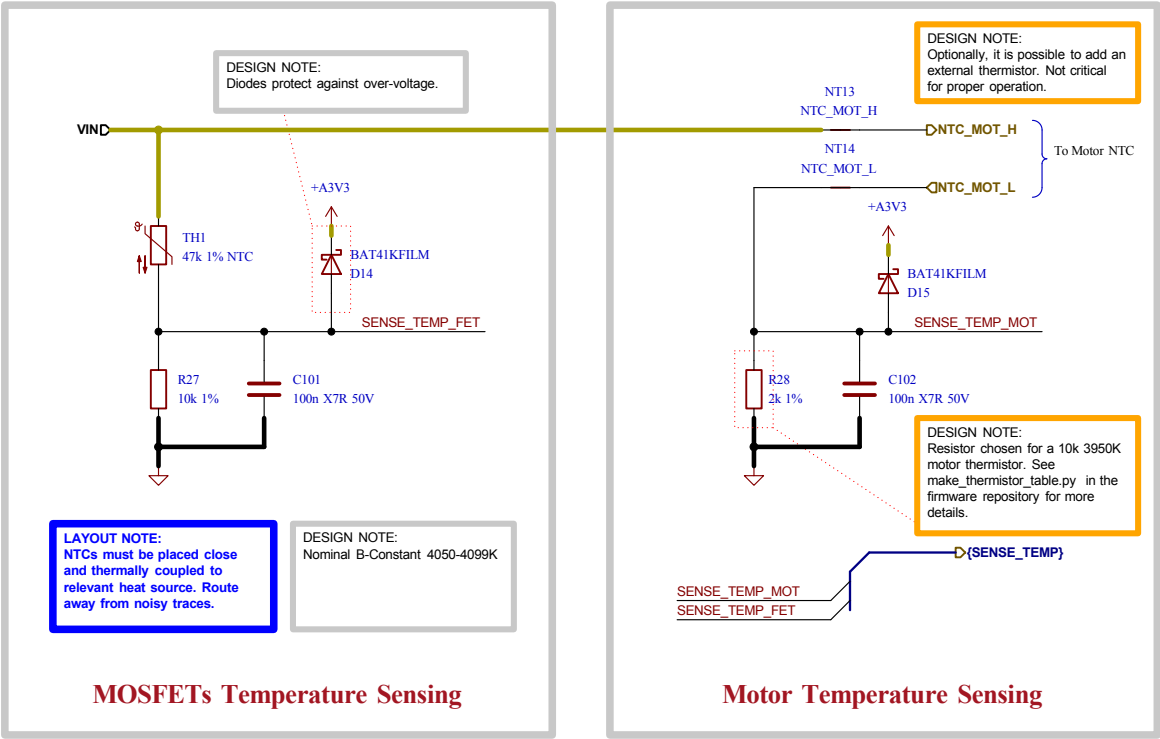
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
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	Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: A4	Sheet: 10 of 21

[11] User - LED Indicators



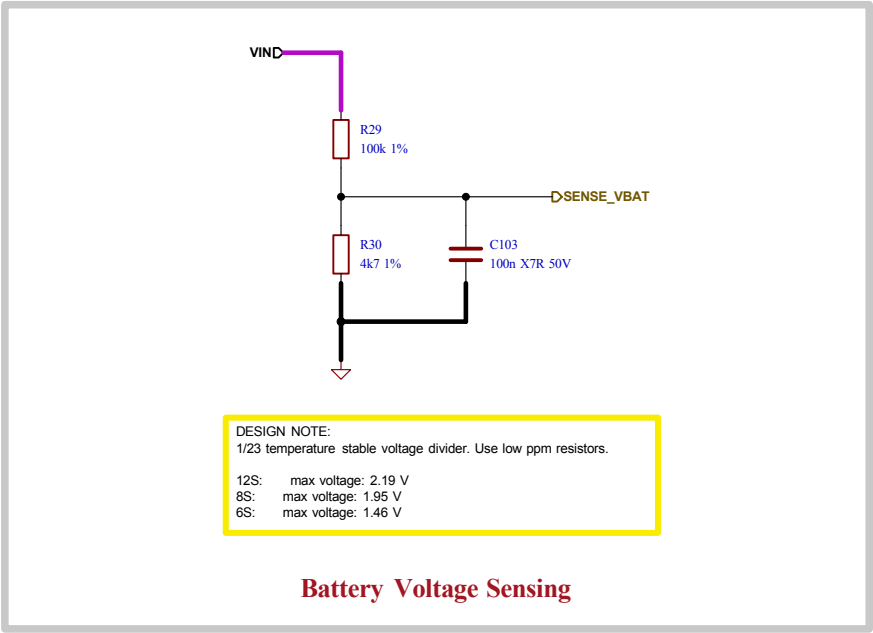
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[12] Sensing - Temperature



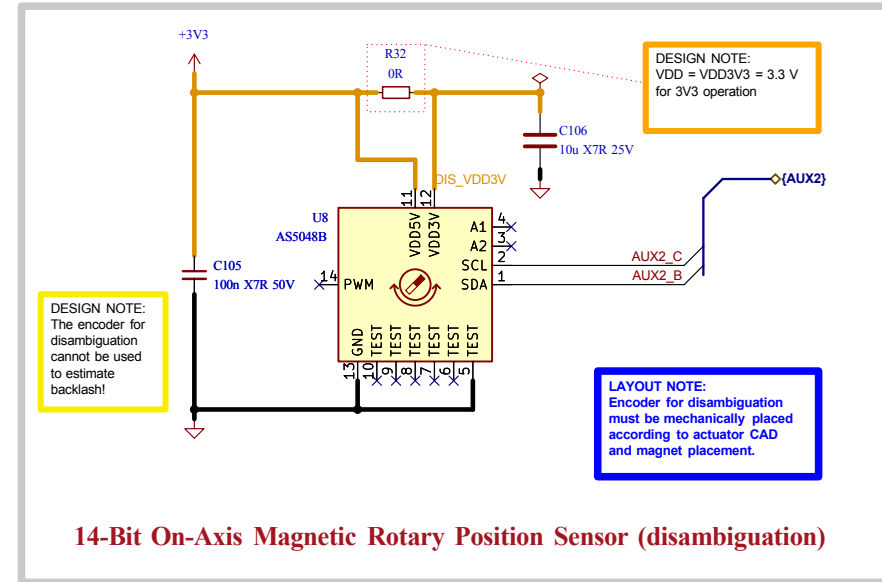
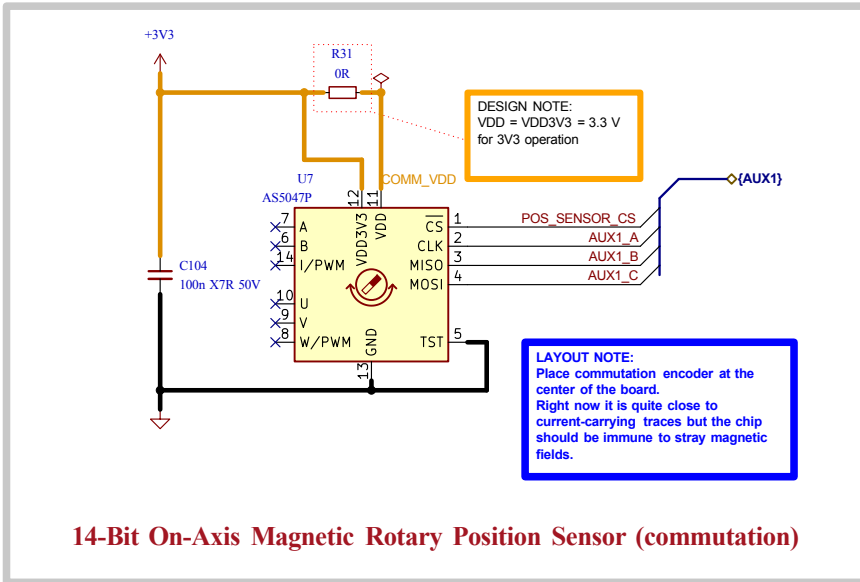
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	Sheet Path: /Project Architecture/Sensing - Temperature/	Reviewer:	Size: A4	Revision: 1.1.1+ (Unreleased)
				Sheet: 12 of 21

[13] Sensing - Battery



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Sensing - Battery	File Name: Sensing - Battery.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14	Revision: 1.1.1+ (Unreleased)
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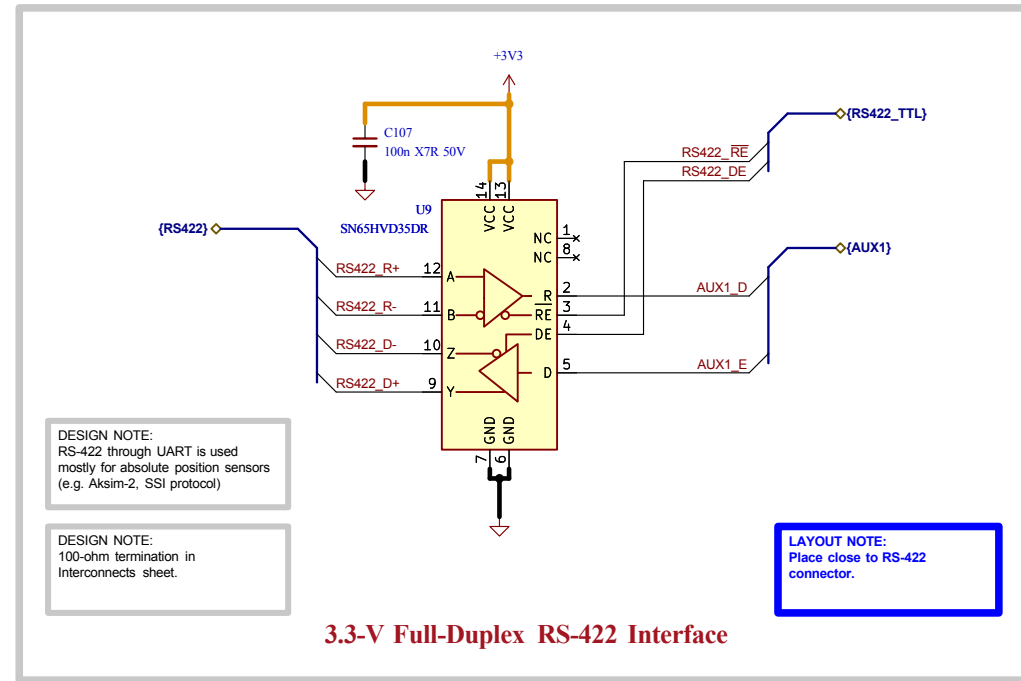
[14] Sensing - Position



DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.

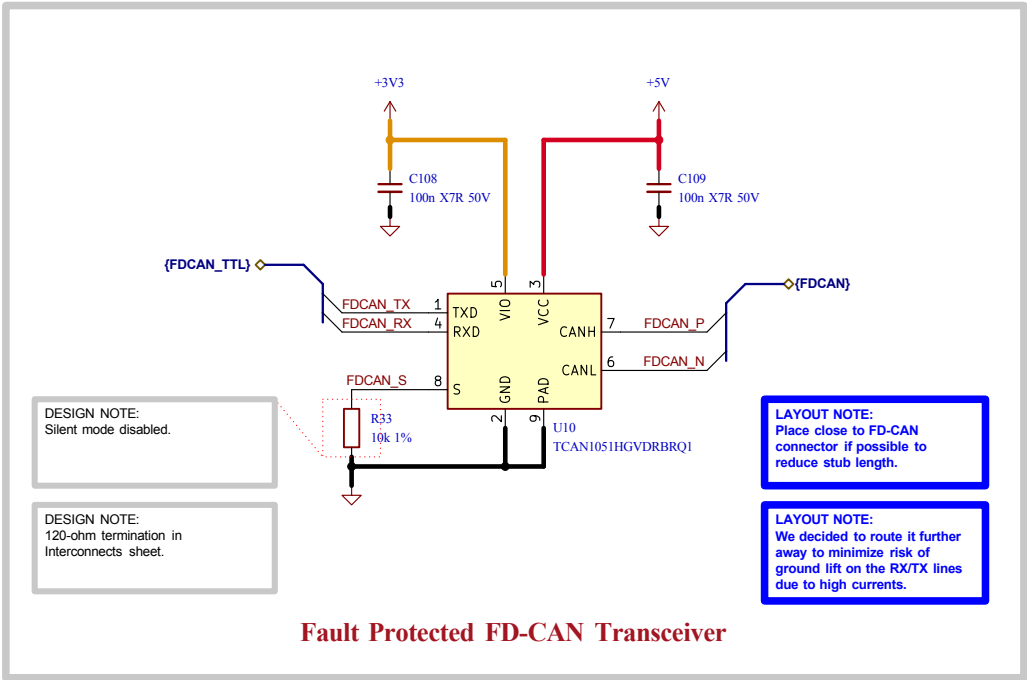
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	Sheet Title: Sensing - Position	Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Path: /Project Architecture/Sensing - Position/	File Name: Sensing - Position.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14
			Reviewer:	Revision: 1.1.1+ (Unreleased)
			Size: A4	Sheet: 14 of 21

[15] Interface - RS-422



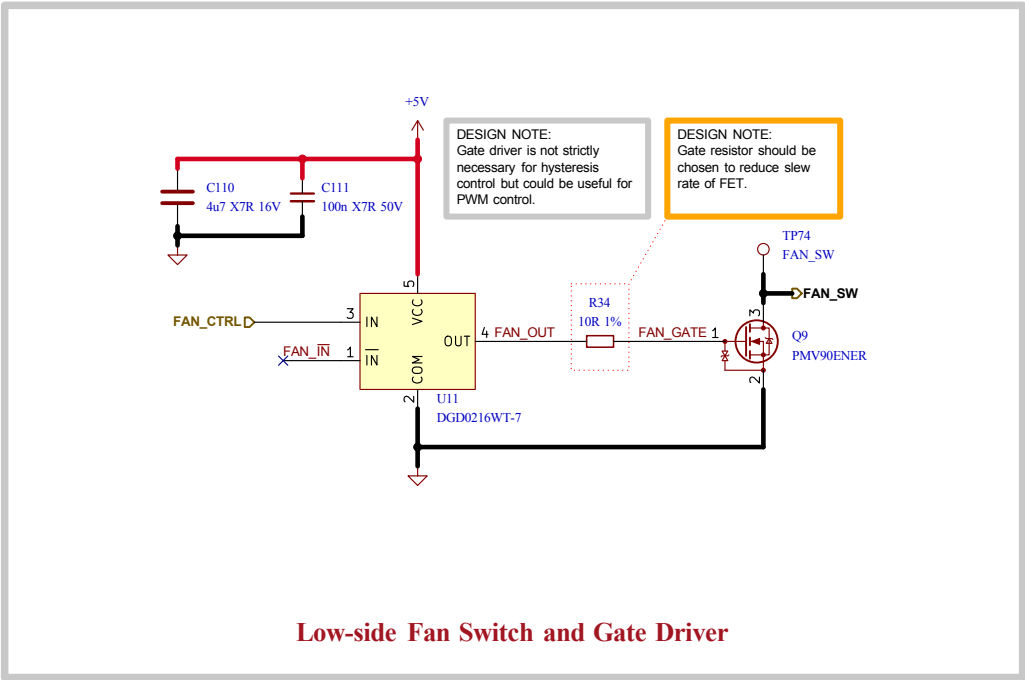
	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - RS-422	File Name: Interface - RS-422.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Interface - RS-422/		Reviewer:	Size: A4	Sheet: 15 of 21

[16] Interface - FD-CAN



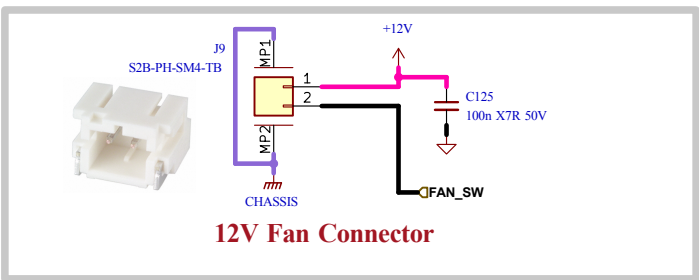
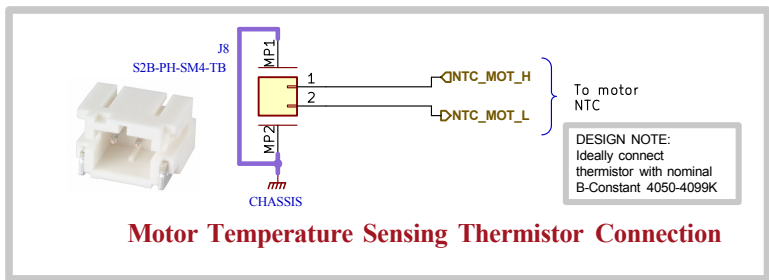
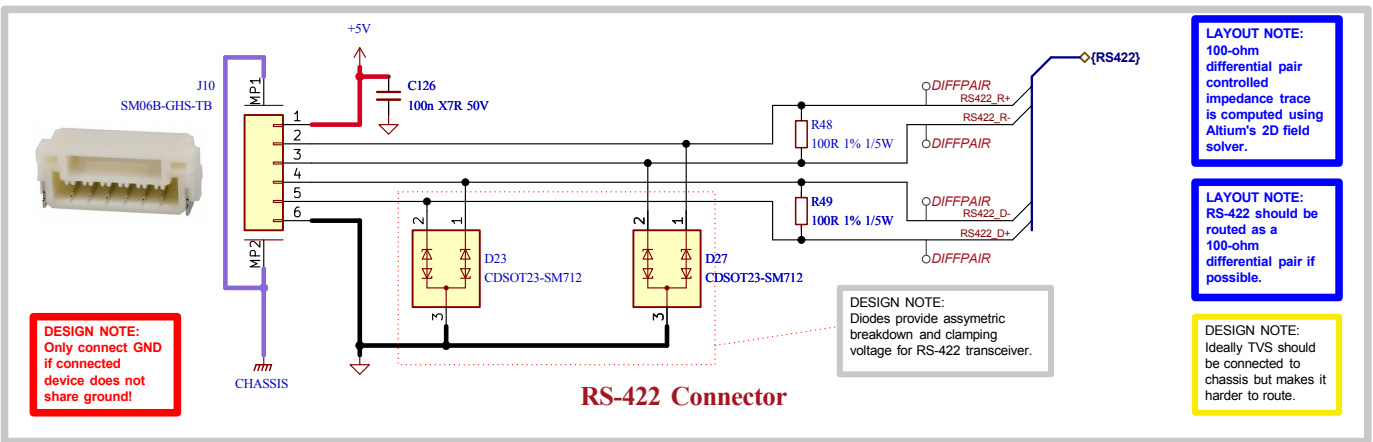
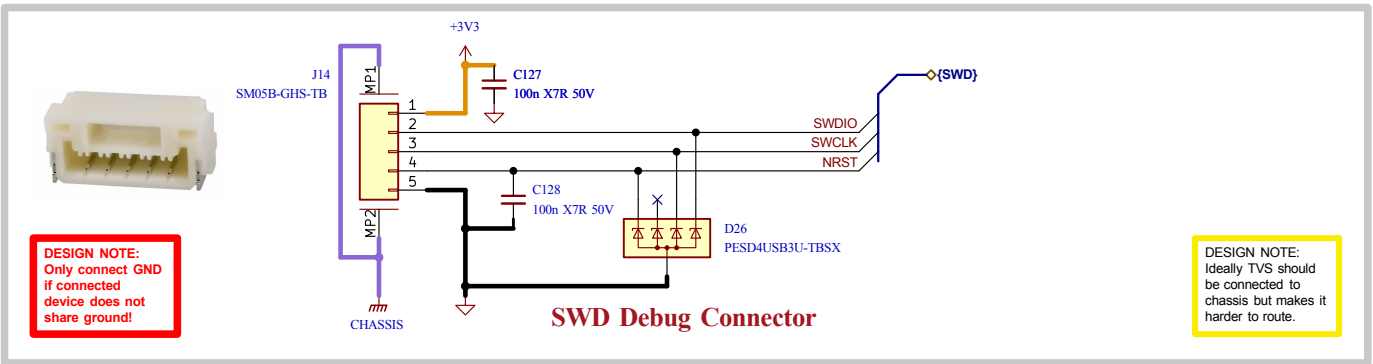
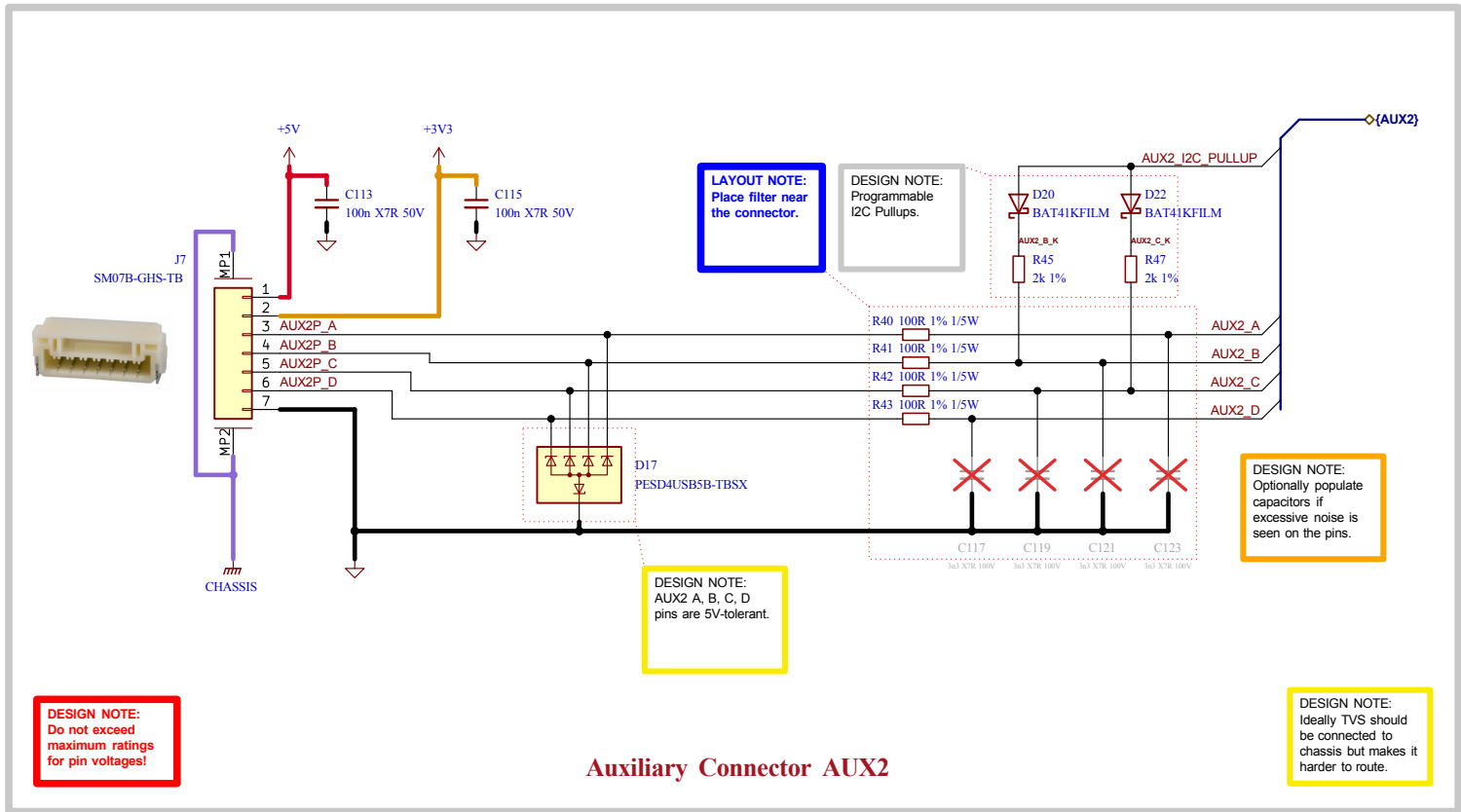
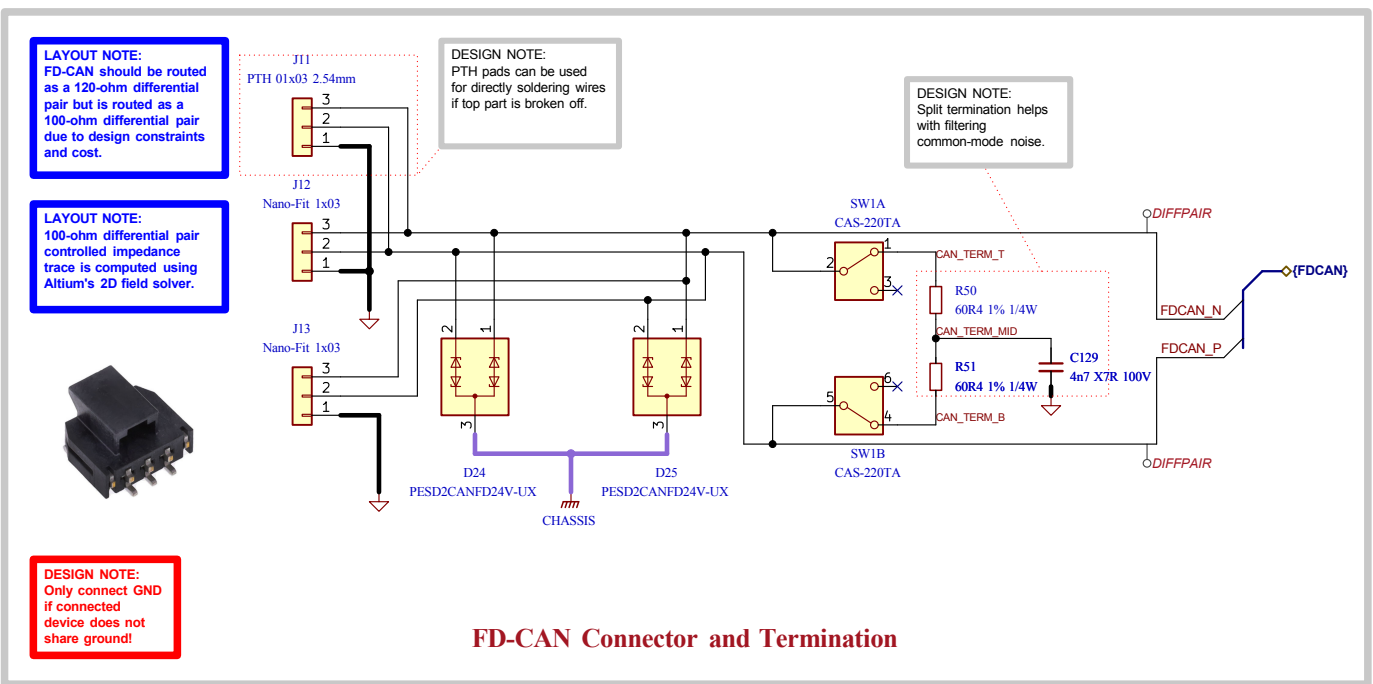
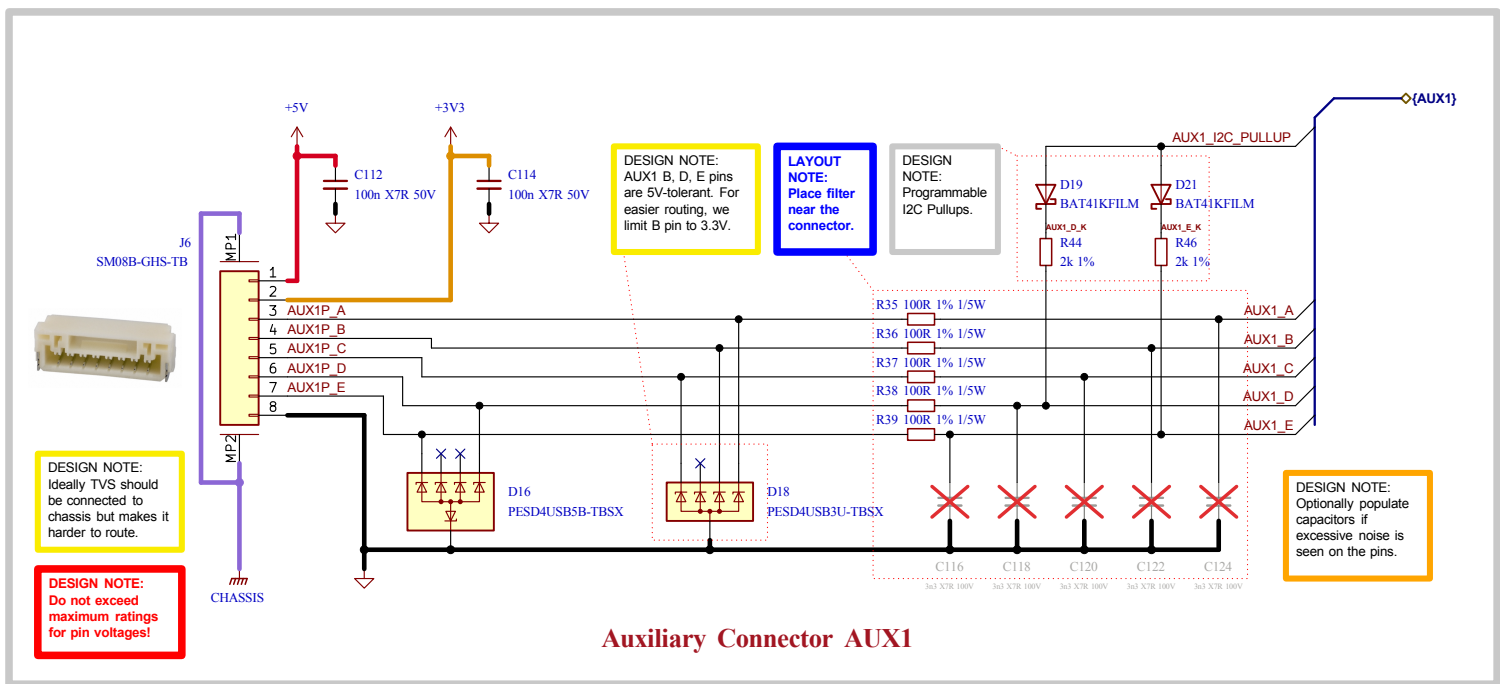
	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - FD-CAN	File Name: Interface - FD-CAN.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 16 of 21

[17] Interface - Fan Control



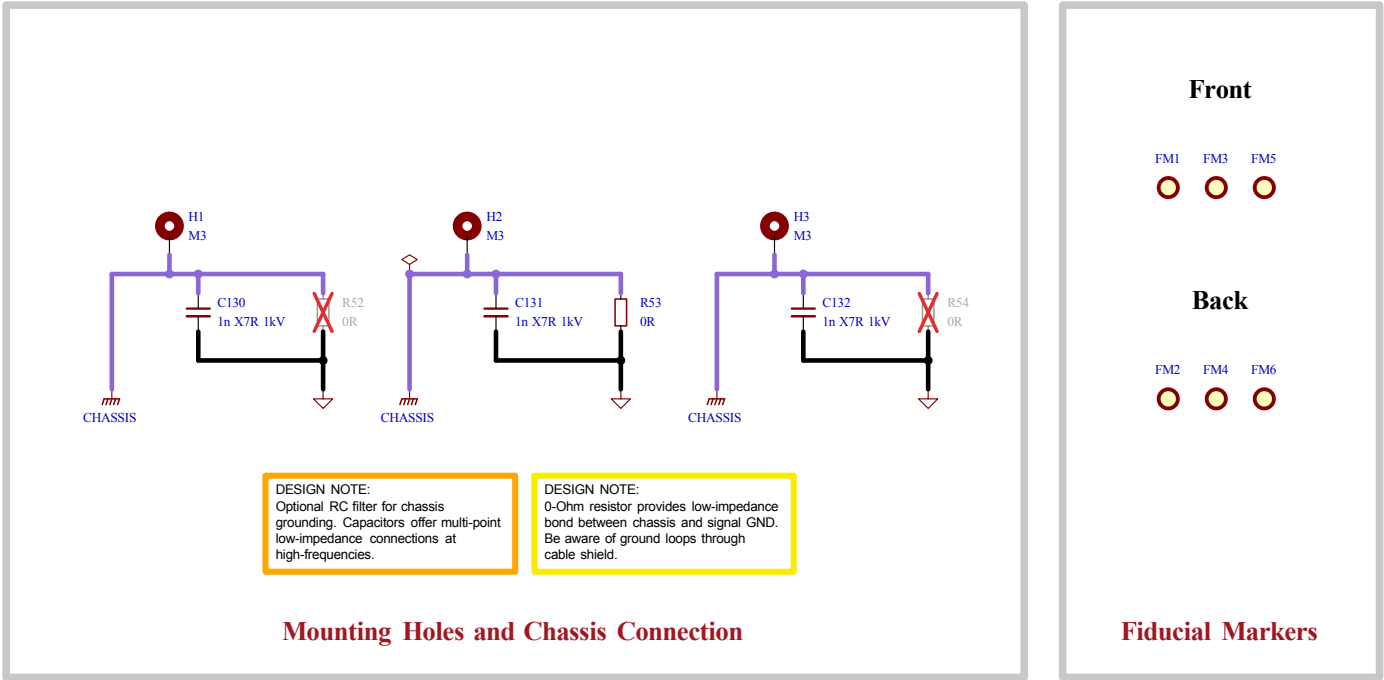
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - Fan Control	File Name: Interface - Fan Control.kicad_sch	Designer: Vincent Nguyen	Date: 2023-11-19	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Interface - Fan Control/		Reviewer:	Size: A4	Sheet: 17 of 21

[18] Interface - Interconnects



	Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A3	Sheet: 18 of 21

[19] Misc - Holes, Fiducials



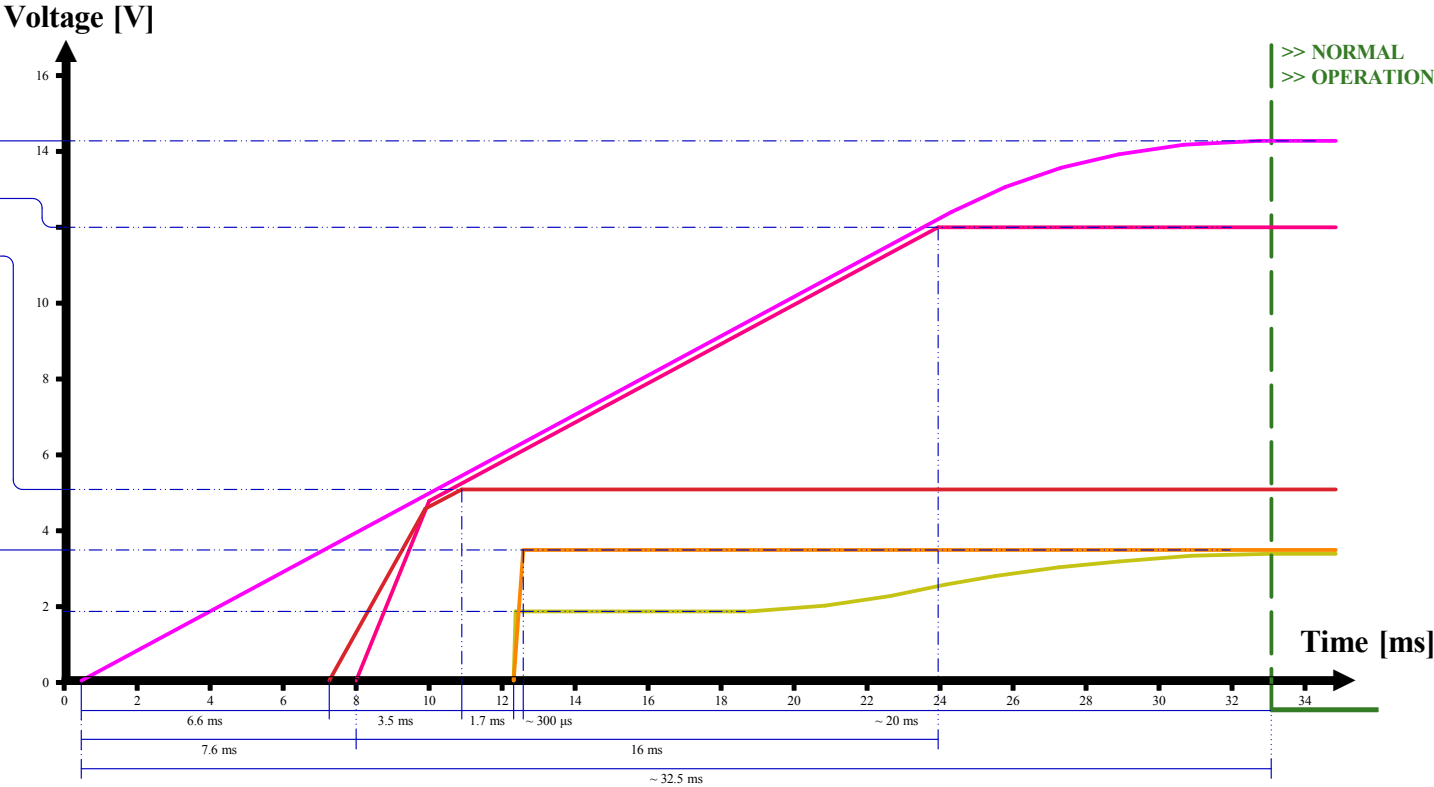
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-22	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 21

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V ± 1.5%
+5V	LMR36506	5V ± 1.5%
+3V3	TPS62172	3.3V ± 3%
+A3V3	LP2992	3.3V ± 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Vincent Nguyen	Date: 2024-03-12	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 20 of 21

[21] Revision History

Version 1.0.0 - 2023-12-12

Added

- TVS protection and termination switch to FD-CAN.
- Low-side switched 12V 600 mA source for external fan.
- LDO for analog supply.
- One TVS diode per power connector.
- Second on-board I2C magnetic encoder for disambiguation.
- ESD protection to all interfaces.
- Over-voltage protection on thermistor ADC inputs.
- Pi filters to inputs of buck regulators and MCU analog supply.
- Decoupling caps next to power pins of connectors.

Changed

- CPH-CPL capacitor to 47 nF (gate driver).
- FD-CAN transceiver IC.
- FETs for top cooled variant.
- Input power TVS diode to bidirectional.
- Moved SOx low-pass filter to MCU section.
- PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- RS422 pinout on connector.
- Buck regulators to optimize for low noise.

Version 1.0.1 - 2024-01-25

Added

- Controller target specifications.
- Credits to moteus on cover page.
- Optional RC-Snubber to power stage.

Fixed

- Chassis guard ring to go around the board.
- CAN and power TVS diodes now go to chassis.
- Clearance between nets to respect IEC60664-1 where possible.
- Comment on precharge.
- Power TVS diode reference designator from "U" to "D".

Changed

- 5V 300 mA buck converter with 600 mA version.
- Chassis-GND capacitor by 1nF 1kV.

Version 1.0.2 - 2024-03-12

Changed

- Power sequencing graph according to experimental data.

Version 1.1.0 - 2024-04-13

Added

- RC snubber passive values.

Fixed

- More vias for VBUS and LMR36006 GND pads.
- Board version voltage reference from +3V3 to +A3V3.

Changed

- Motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

Version 1.1.1 - 2025-01-18

Fixed

- Replace non-stocked components.

	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Revision History	File Name: Revision_History.kicad_sch	Designer: Vincent Nguyen	Date: 2024-01-03	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 21 of 21